Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP(01CT1513)	**Aim: History Equalization & Matching Techniques**	
Open-ended-Problem: 08	**Date:22-08-2026**	**Enrollment No:92400133121**

AIM:

- Five different degraded images are provided.
- Apply histogram equalization and histogram matching techniques to enhance the visual quality
- After processing, describe observations in detail for every enhancement step
- Explain how each technique changes the image contrast and clarity.
- Conclude with a summary of which approach worked better for each image.

Theory:

Histogram:

- A histogram is a graphical representation that shows the distribution of pixel intensities in an image.
- It displays the frequency of each intensity value on the horizontal axis and the number of pixels with that intensity value on the vertical axis.

Histogram Equalization:

- Histogram equalization is a technique used to enhance the contrast of an image.
- It stretches the histogram of an image to cover the full range of intensity values, resulting in a more balanced and visually appealing image.
- In this experiment the input image is first converted into grayscale image then equalizeHist() function is used to perform histogram equalization.

Histogram Matching:

- Histogram matching is a process of transforming an image to match the histogram of a reference image.

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- It is used to transfer the characteristics of the reference image's histogram to the target image, making them visually similar in terms of intensity distribution.
- Unlike histogram equalization which automatically redistributes intensities, histogram matching allows the enhanced image to follow the intensity characteristics of reference image.
- So, this method can overall adjust the contrast according to reference image, provide more controlled visual appearance and improve the brightness and tonal distribution.
- In this experiment the input image is first converted into gray scale image then matchHistogram() function is used to perform histogram matching.

Flowchart:

The flowchart for the program will consist of the following steps:

- Load the input image.
- Generate and display the histogram of the image.
- Apply histogram equalization to enhance the contrast.
- Generate and display the histogram of the equalized image.
- Load the reference image for histogram matching.
- Perform histogram matching between the input image and reference image.
- Generate and display the histogram of the matched image.
- Display the original image, equalized image, and matched image side by side.
- Save the equalized and matched images (optional).

MAIN FUNCTIONS:

Code:

```
pip install scikit-image
pip install opencv-python
```

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
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```
#import libraries
#cv2 lib reads in this format : blue-green-red plane (bgr)

import cv2
import numpy as np
import matplotlib.pyplot as plt
from skimage.exposure import match_histograms

def generate_histogram(image):

    histogram = cv2.calcHist(
        [image],
        [0],
        None,
        [256],
        [0, 256])

    return histogram

def perform_histogram_equalization(image):

  gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
  equalized_image = cv2.equalizeHist(gray_image) # Apply histogram equalization

  return equalized_image

def perform_histogram_matching(input_image, reference_image):

  gray_input = cv2.cvtColor(input_image, cv2.COLOR_BGR2GRAY)

  gray_reference = cv2.cvtColor(reference_image, cv2.COLOR_BGR2GRAY)

  input_hist = generate_histogram(gray_input) # Calculate histograms
  reference_hist = generate_histogram(gray_reference)
```

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```
  matched_image = match_histograms(gray_input, gray_reference)# Perform histogram
matching
  matched_image = np.uint8(matched_image)

  return matched_image
```

FIRST IMAGE:



PROBLEM:

- It is a low contrast image.
- The image details are not visible clearly.

Code:

```
# Load image 1

input_path = 'images/Picture1.png'
input_image = cv2.imread(input_path)

# Generate and display the histogram of the input image

gray_input = cv2.cvtColor(input_image,cv2.COLOR_BGR2GRAY)
input_hist = generate_histogram(gray_input)


# Histogram equalization
```

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
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Open-ended-Problem: 08	**Date:22-08-2026**	**Enrollment No:92400133121**

```
equalized_image = perform_histogram_equalization(input_image)

equalized_hist = generate_histogram(equalized_image)

# reference image for histogram matching
reference_path = 'images/images.png'
reference_image = cv2.imread(reference_path)

# Perform histogram matching

matched_image = perform_histogram_matching(input_image, reference_image)
matched_hist = generate_histogram(matched_image)


plt.figure(figsize=(15,5))

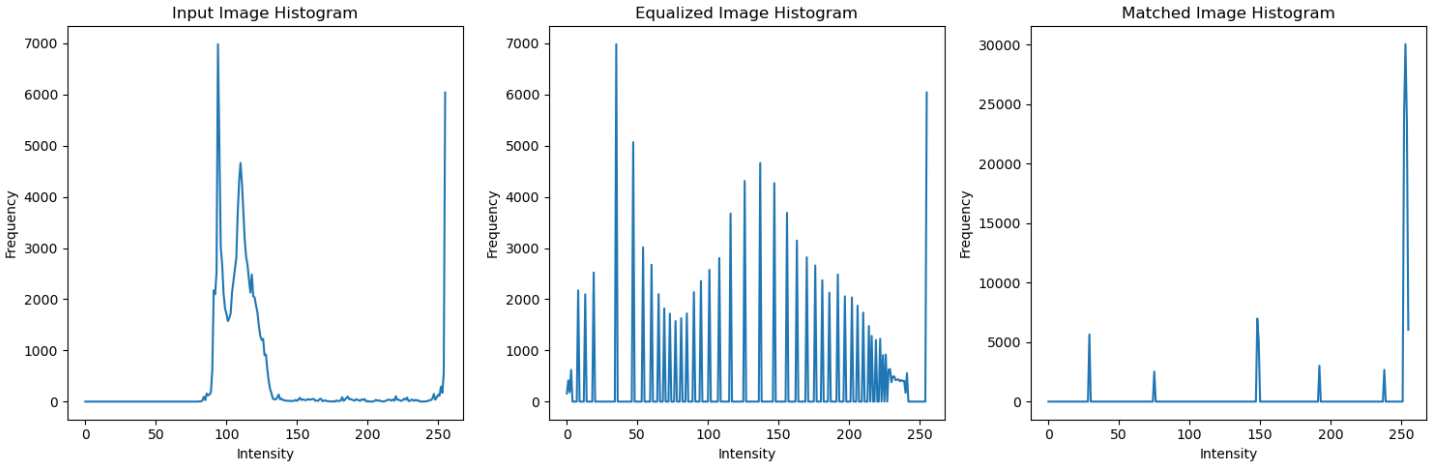
plt.subplot(1,3,1)
plt.plot(input_hist)
plt.title("Input Image Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")

plt.subplot(1,3,2)
plt.plot(equalized_hist)
plt.title("Equalized Image Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")

plt.subplot(1,3,3)
plt.plot(matched_hist)
plt.title("Matched Image Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()
```

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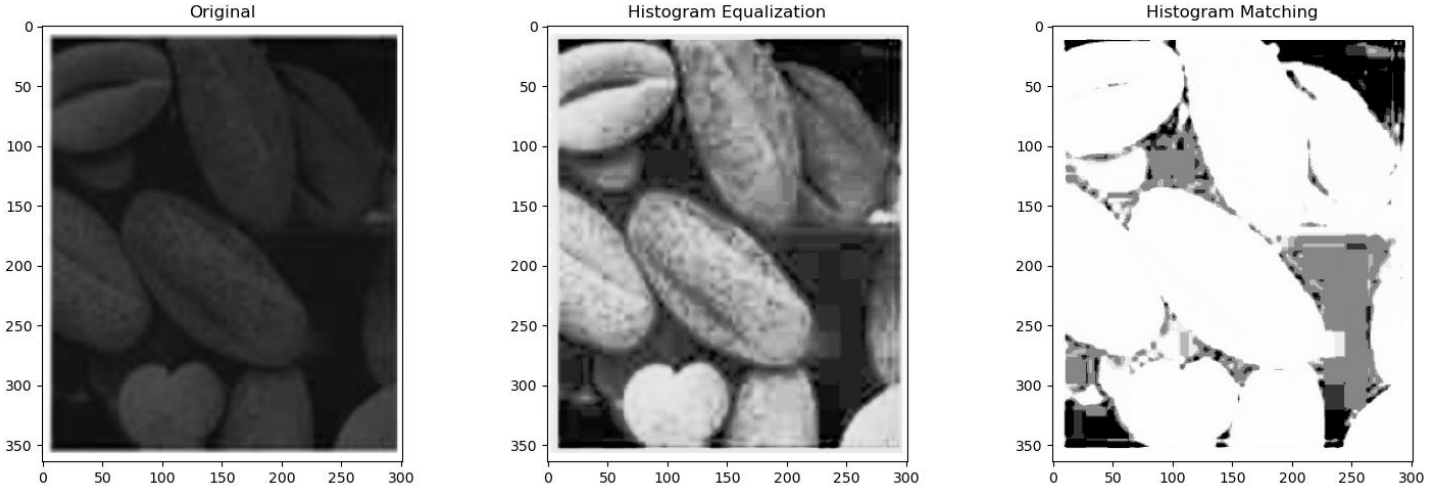

```
plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
plt.imshow(gray_input,cmap='gray')
plt.title("Original")

plt.subplot(1,3,2)
plt.imshow(equalized_image,cmap='gray')
plt.title("Histogram Equalization")

plt.subplot(1,3,3)
plt.imshow(matched_image,cmap='gray')
plt.title("Histogram Matching")

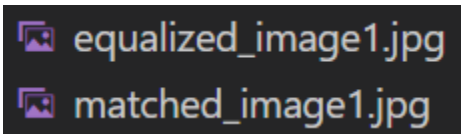
plt.tight_layout()
plt.show()
```



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Open-ended-Problem: 08	**Date:22-08-2026**	**Enrollment No:92400133121**

```
# Save the equalized and matched images
equalized_path = 'equalized_image1.jpg'
matched_path = 'matched_image1.jpg'

cv2.imwrite(equalized_path, equalized_image)
cv2.imwrite(matched_path, matched_image)
```



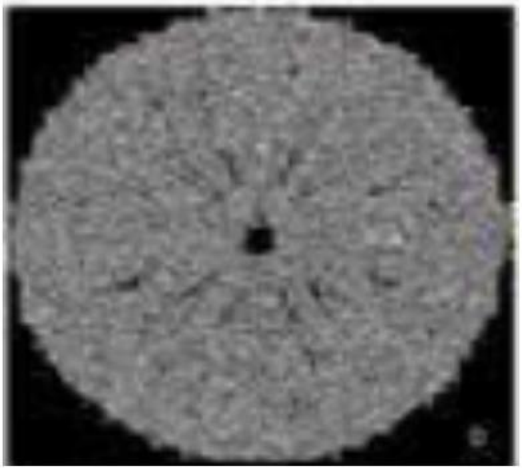
Observation:

- Histogram equalization improves the contrast and makes the objects and their textures more visible. However, some regions become excessively bright or dark.
- Histogram matching significantly increases the brightness. But large portions of the image become nearly white because of which some internal details are lost.

Conclusion

Histogram equalization performs better than histogram matching because it improves contrast while preserving more object details.

SECOND IMAGE:



PROBLEM:

- Given image has poor visual contrast.

- Some internal structures are difficult to distinguish.

Code:

```
# Load image 2

input_path = 'images/Picture2.png'
input_image = cv2.imread(input_path)

# Generate and display the histogram of the input image

gray_input = cv2.cvtColor(input_image,cv2.COLOR_BGR2GRAY)
input_hist = generate_histogram(gray_input)


# Histogram equalization

equalized_image = perform_histogram_equalization(input_image)

equalized_hist = generate_histogram(equalized_image)

# reference image for histogram matching
reference_path = 'images/images.png'
reference_image = cv2.imread(reference_path)

# Perform histogram matching

matched_image = perform_histogram_matching(input_image, reference_image)
matched_hist = generate_histogram(matched_image)


plt.figure(figsize=(15,5))

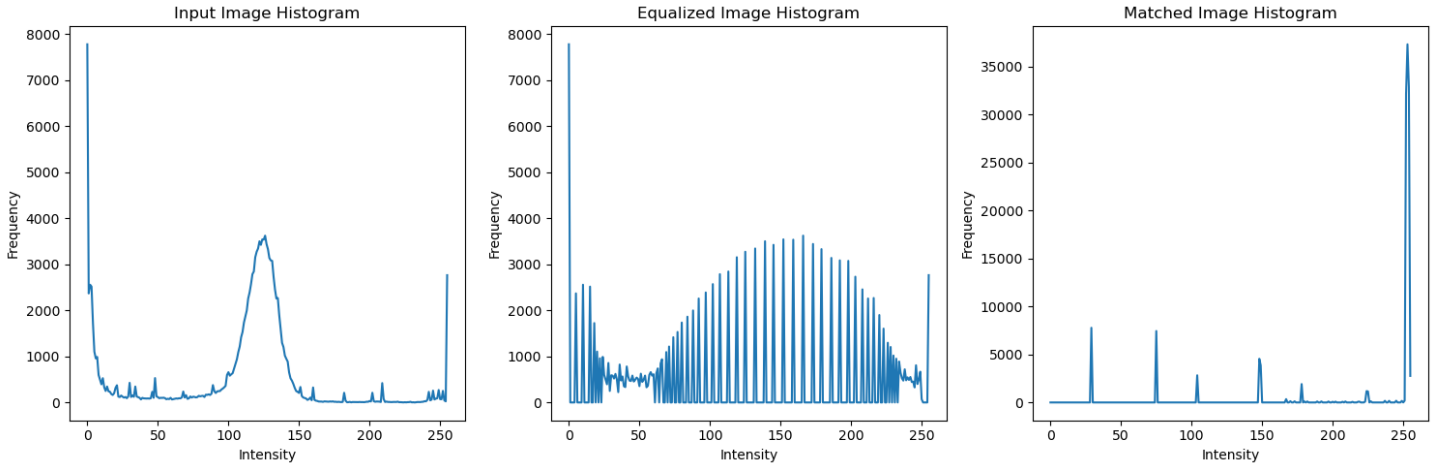
plt.subplot(1,3,1)
plt.plot(input_hist)
plt.title("Input Image Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")

plt.subplot(1,3,2)
plt.plot(equalized_hist)
```

```
plt.title("Equalized Image Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")

plt.subplot(1,3,3)
plt.plot(matched_hist)
plt.title("Matched Image Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()
```




```
plt.figure(figsize=(15,5))

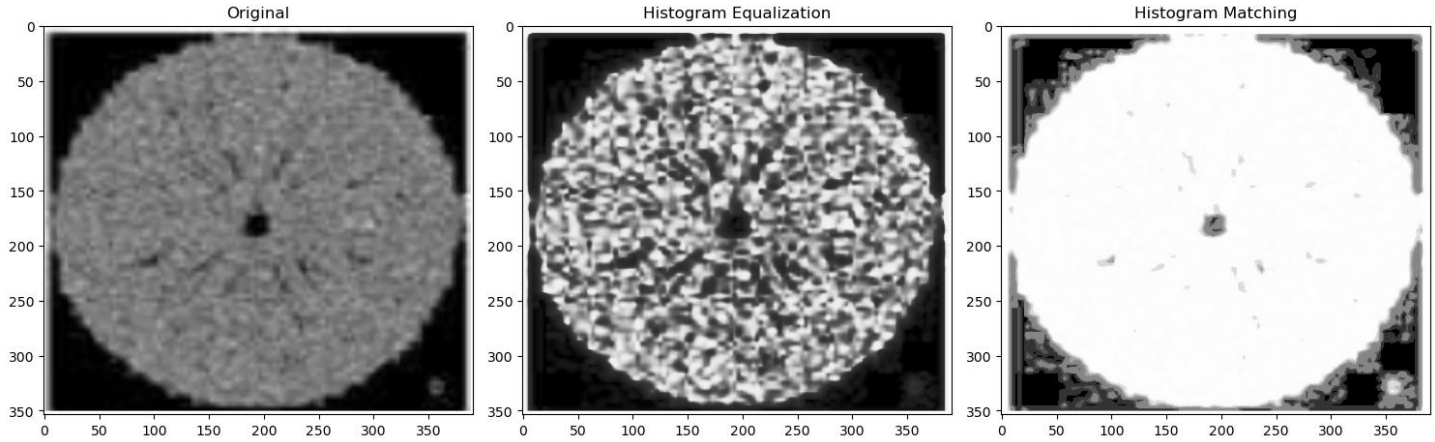
plt.subplot(1,3,1)
plt.imshow(gray_input,cmap='gray')
plt.title("Original")

plt.subplot(1,3,2)
plt.imshow(equalized_image,cmap='gray')
plt.title("Histogram Equalization")

plt.subplot(1,3,3)
plt.imshow(matched_image,cmap='gray')
plt.title("Histogram Matching")

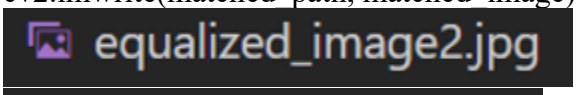
plt.tight_layout()
plt.show()
```

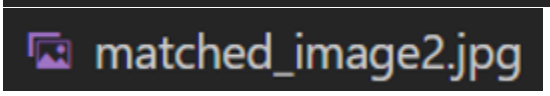
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Open-ended-Problem: 08	**Date:22-08-2026**	**Enrollment No:92400133121**




```
# Save the equalized and matched images (optional)
equalized_path = 'equalized_image2.jpg'
matched_path = 'matched_image2.jpg'

cv2.imwrite(equalized_path, equalized_image)
cv2.imwrite(matched_path, matched_image)
```





Observation

- Histogram equalization enhances the internal structures and makes intensity variations more prominent. However, noise in the image is also amplified.
- Histogram matching makes the image brighter
  But large regions become nearly white, reducing the visibility of internal details.

Conclusion

Histogram equalization performs better than histogram matching because it reveals more internal structures and preserves more useful intensity variations, although some noise is amplified.

THIRD IMAGE:

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Open-ended-Problem: 08	**Date:22-08-2026**	**Enrollment No:92400133121**



PROBLEM:

- The given image is very dark image.
- It is little bit difficult to distinguish the object from background.
- Object details are not clearly visible.

Code:

```
# Load image 3

input_path = 'images/Picture3.png'
input_image = cv2.imread(input_path)

# Generate and display the histogram of the input image

gray_input = cv2.cvtColor(input_image,cv2.COLOR_BGR2GRAY)
input_hist = generate_histogram(gray_input)


# Histogram equalization

equalized_image = perform_histogram_equalization(input_image)

equalized_hist = generate_histogram(equalized_image)

# reference image for histogram matching
reference_path = 'images/images.png'
reference_image = cv2.imread(reference_path)

# Perform histogram matching
```

```
matched_image = perform_histogram_matching(input_image, reference_image)
matched_hist = generate_histogram(matched_image)

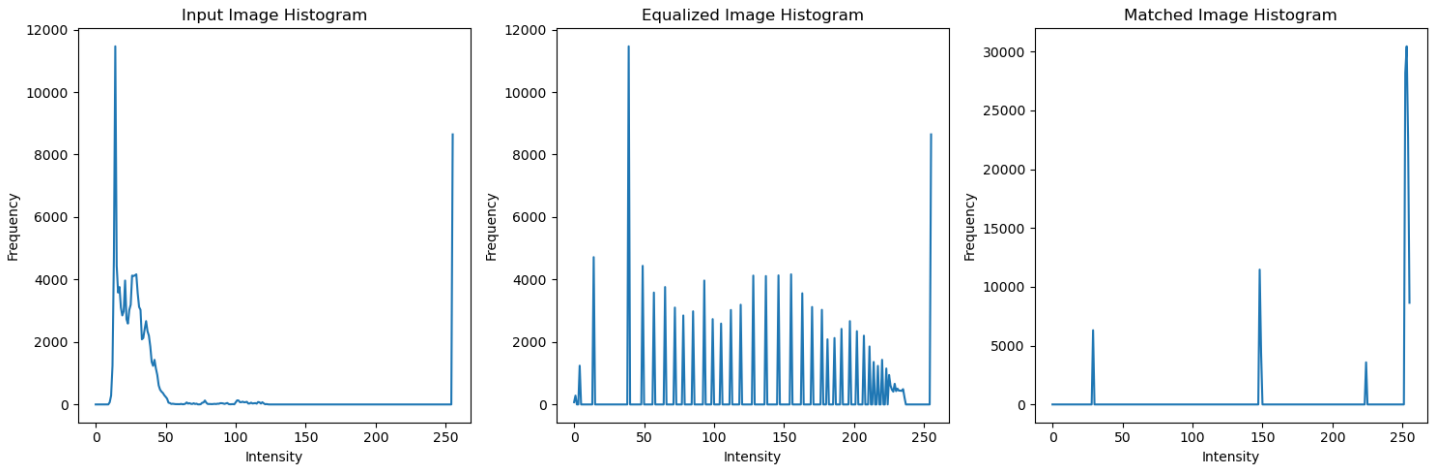

plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
plt.plot(input_hist)
plt.title("Input Image Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")

plt.subplot(1,3,2)
plt.plot(equalized_hist)
plt.title("Equalized Image Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")

plt.subplot(1,3,3)
plt.plot(matched_hist)
plt.title("Matched Image Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()
```




```
plt.figure(figsize=(15,5))
```

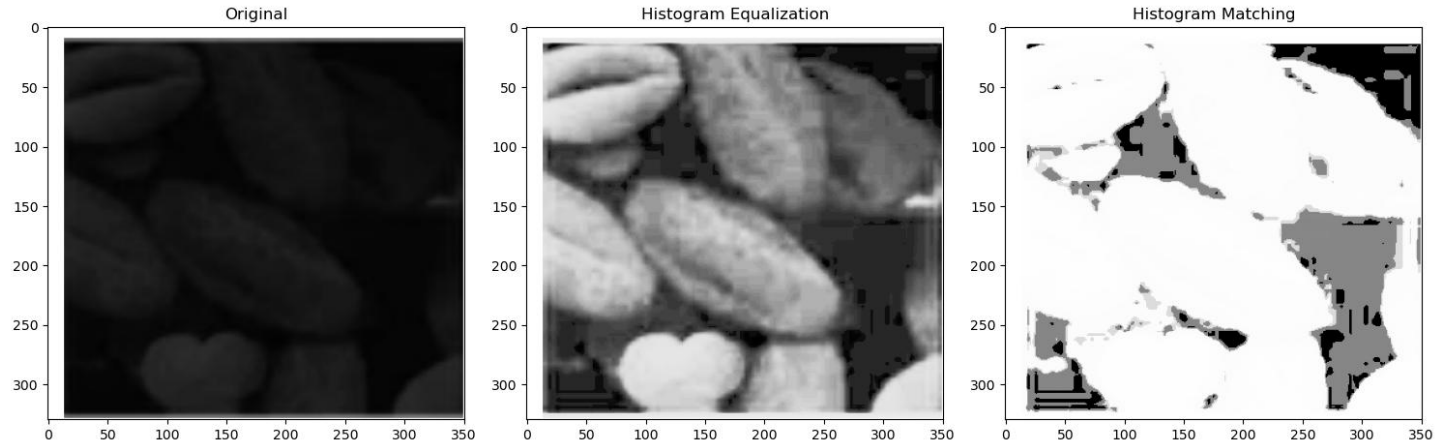
Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP(01CT1513)	**Aim: History Equalization & Matching Techniques**	
Open-ended-Problem: 08	**Date:22-08-2026**	**Enrollment No:92400133121**

```
plt.subplot(1,3,1)
plt.imshow(gray_input,cmap='gray')
plt.title("Original")

plt.subplot(1,3,2)
plt.imshow(equalized_image,cmap='gray')
plt.title("Histogram Equalization")

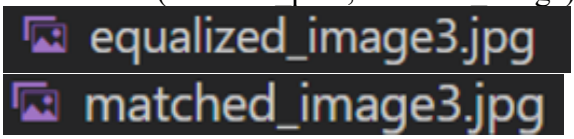
plt.subplot(1,3,3)
plt.imshow(matched_image,cmap='gray')
plt.title("Histogram Matching")

plt.tight_layout()
plt.show()
```




```
# Save the equalized and matched images
equalized_path = 'equalized_image3.jpg'
matched_path = 'matched_image3.jpg'

cv2.imwrite(equalized_path, equalized_image)
cv2.imwrite(matched_path, matched_image)
```



Observation

- Histogram equalization significantly improves brightness and contrast, revealing the shapes, boundaries, and textures of the objects.

- Histogram matching produces excessive brightness, causing several regions to appear nearly white and some internal textures and details are lost.

Conclusion

Histogram equalization performs better than histogram matching because it provides stronger visibility of the objects while retaining more details.

FOURTH IMAGE:



PROBLEM:

- The input image is severely underexposed
- The surrounding details are barely visible

Code:

```
# Load image 4

input_path = 'images/Picture4.png'
input_image = cv2.imread(input_path)

# Generate and display the histogram of the input image

gray_input = cv2.cvtColor(input_image,cv2.COLOR_BGR2GRAY)
input_hist = generate_histogram(gray_input)


# Histogram equalization

equalized_image = perform_histogram_equalization(input_image)
```

```
equalized_hist = generate_histogram(equalized_image)

# reference image for histogram matching
reference_path = 'images/images.png'
reference_image = cv2.imread(reference_path)

# Perform histogram matching

matched_image = perform_histogram_matching(input_image, reference_image)
matched_hist = generate_histogram(matched_image)


plt.figure(figsize=(15,5))

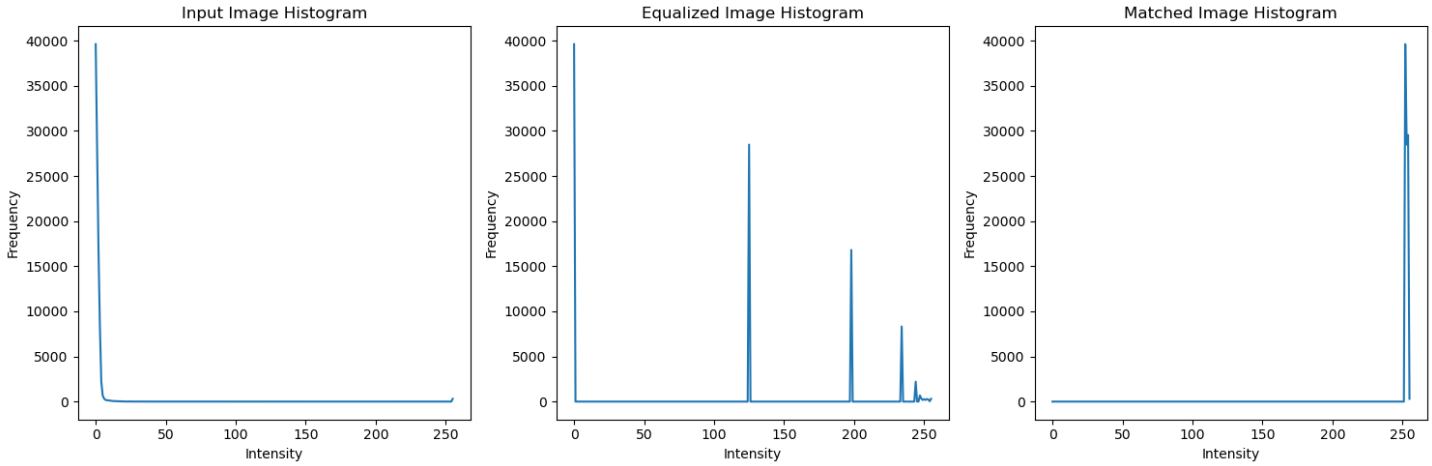
plt.subplot(1,3,1)
plt.plot(input_hist)
plt.title("Input Image Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")

plt.subplot(1,3,2)
plt.plot(equalized_hist)
plt.title("Equalized Image Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")

plt.subplot(1,3,3)
plt.plot(matched_hist)
plt.title("Matched Image Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()
```

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP(01CT1513)	**Aim: History Equalization & Matching Techniques**	
Open-ended-Problem: 08	**Date:22-08-2026**	**Enrollment No:92400133121**




```
plt.figure(figsize=(15,5))

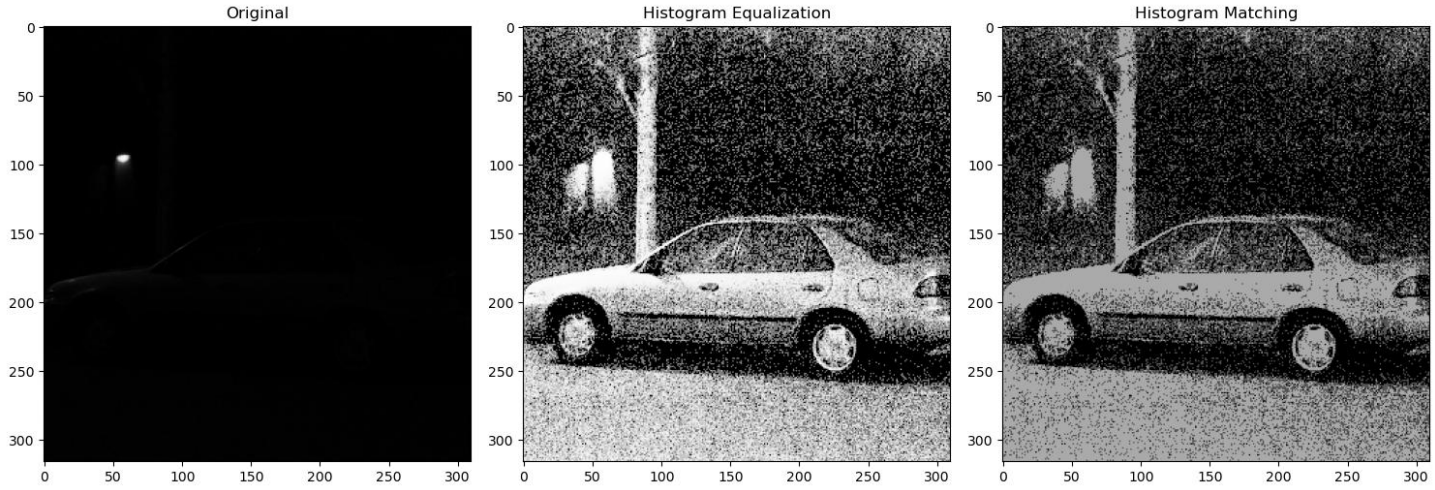
plt.subplot(1,3,1)
plt.imshow(gray_input,cmap='gray')
plt.title("Original")

plt.subplot(1,3,2)
plt.imshow(equalized_image,cmap='gray')
plt.title("Histogram Equalization")

plt.subplot(1,3,3)
plt.imshow(matched_image,cmap='gray')
plt.title("Histogram Matching")

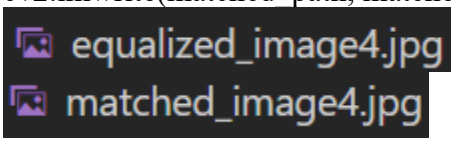
plt.tight_layout()
plt.show()
```

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Open-ended-Problem: 08	**Date:22-08-2026**	**Enrollment No:92400133121**




```
# Save the equalized and matched images
equalized_path = 'equalized_image4.jpg'
matched_path = 'matched_image4.jpg'

cv2.imwrite(equalized_path, equalized_image)
cv2.imwrite(matched_path, matched_image)
```



Observation

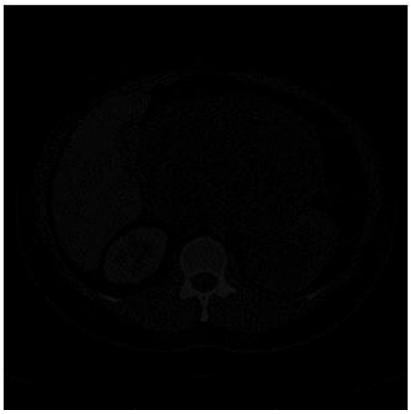
- Histogram equalization improves visibility and reveals the car, wheels, windows, road, and background. However, it also significantly amplifies image noise.
- Histogram matching improves the visibility of the car and its surroundings. However, noise remains in the enhanced image.

Conclusion

Histogram equalization performs better than history matching because it reveals more details from the severely dark image, although it produces some noise

FIFTH IMAGE:

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Open-ended-Problem: 08	**Date:22-08-2026**	**Enrollment No:92400133121**



PROBLEM:

- The image is very dark which almost looks blank.
- Due to its poor intensity distribution the boundaries, fine details nothing are visible
- One cannot identify the internal structures in the given image.

Code:

```
# Load image 5

input_path = 'images/Picture5.png'
input_image = cv2.imread(input_path)

# Generate and display the histogram of the input image

gray_input = cv2.cvtColor(input_image,cv2.COLOR_BGR2GRAY)
input_hist = generate_histogram(gray_input)


# Histogram equalization

equalized_image = perform_histogram_equalization(input_image)

equalized_hist = generate_histogram(equalized_image)

# reference image for histogram matching
reference_path = 'images/images.png'
reference_image = cv2.imread(reference_path)

# Perform histogram matching
```

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Open-ended-Problem: 08	**Date:22-08-2026**	**Enrollment No:92400133121**

```
matched_image = perform_histogram_matching(input_image, reference_image)
matched_hist = generate_histogram(matched_image)

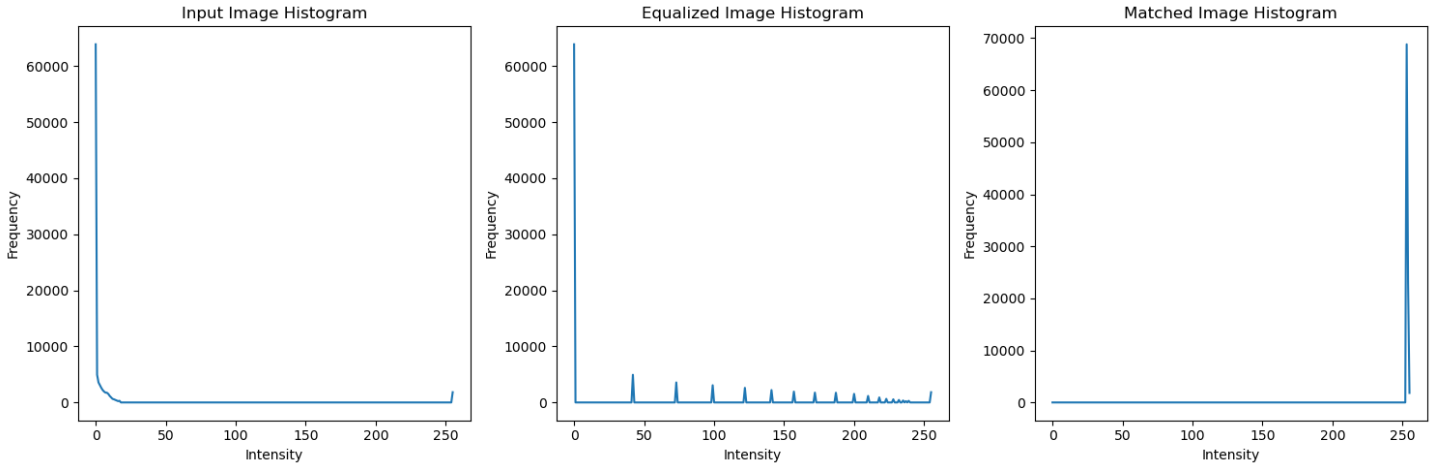

plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
plt.plot(input_hist)
plt.title("Input Image Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")

plt.subplot(1,3,2)
plt.plot(equalized_hist)
plt.title("Equalized Image Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")

plt.subplot(1,3,3)
plt.plot(matched_hist)
plt.title("Matched Image Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()
```




```
plt.figure(figsize=(15,5))

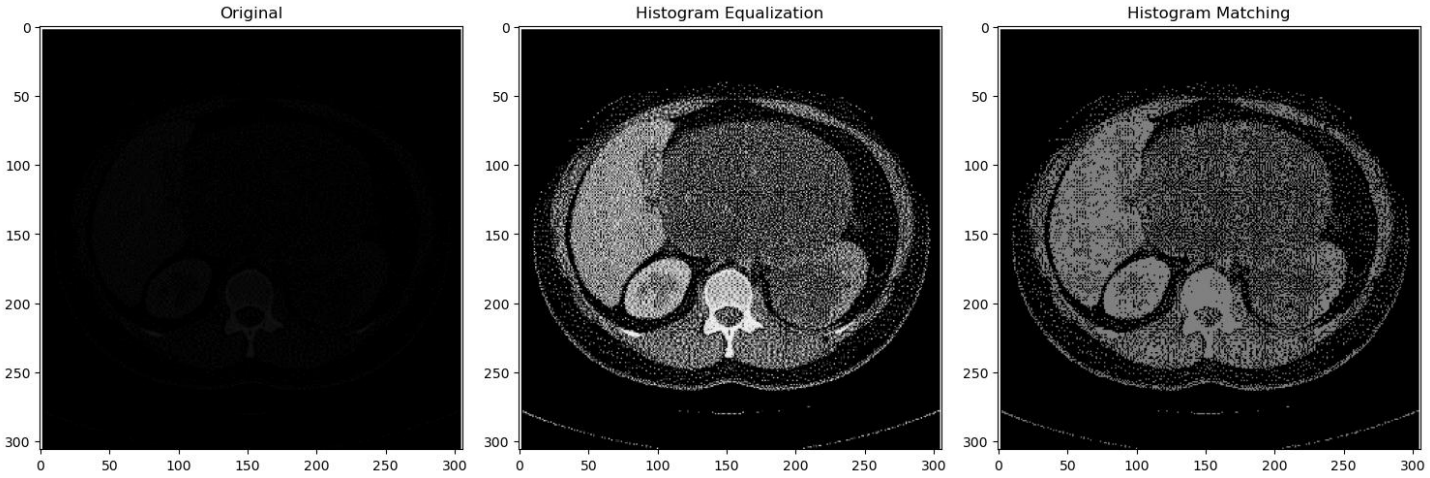
plt.subplot(1,3,1)
```

```
plt.imshow(gray_input,cmap='gray')
plt.title("Original")

plt.subplot(1,3,2)
plt.imshow(equalized_image,cmap='gray')
plt.title("Histogram Equalization")

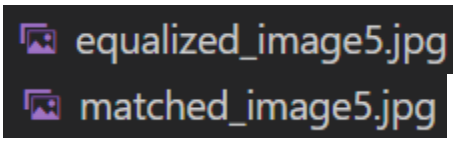
plt.subplot(1,3,3)
plt.imshow(matched_image,cmap='gray')
plt.title("Histogram Matching")

plt.tight_layout()
plt.show()
```




```
# Save the equalized and matched images
equalized_path = 'equalized_image5.jpg'
matched_path = 'matched_image5.jpg'

cv2.imwrite(equalized_path, equalized_image)
cv2.imwrite(matched_path, matched_image)
```



Observation

- Histogram equalization greatly enhances the contrast and reveals the internal structures, but it also strongly amplifies the noise.

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
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Open-ended-Problem: 08	**Date:22-08-2026**	**Enrollment No:92400133121**

- Histogram matching improves the visibility of the anatomical structures and some fine structural details are less distinguishable.

Conclusion:

Histogram equalization performs better than histogram matching because it provides stronger contrast enhancement and reveals more anatomical details, despite some amplification of noise.

Final Conclusion:

- ✓ Histogram equalization performed better for all five images by improving contrast and revealing more details.
- ✓ Although it amplified some noise, it provided better visual enhancement than histogram matching for the given images.